
Identification of the Causative Agent of Emerging Respiratory Epidemic (SPOK-24)

Technical brief

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Subject: Confirmation of a Novel Zoonotic Spooky-like Virus 2024 (SPOK-24)

1. Executive Summary

The World Health Organization, in collaboration with the Australian Department of Health and Aged Care and global reference laboratories, has confirmed that the causative agent responsible for the ongoing respiratory epidemic originally identified in Brisbane, Australia, is a novel paramyxovirus related to Hendra virus, tentatively designated:

- Virus Name (provisional): Spooky-like Virus 2024 (SPOK-24)
- Disease Name (provisional): Spooky-Associated Respiratory Syndrome 2024 (SPOKARS-24)

This newly identified virus, SPOK-24, is genetically distinct from known Spookyvirus strains, with adaptations that have resulted in lower virulence, broader human tissue tropism, and sustained human-to-human transmissibility.

2. Virological Characteristics

- Virus Family: Spookyviridae
- Genus: Spookyvirus
- Provisional Species Name: Spookyvirus australisensis (pending ICTV review)
- Genome: Enveloped, negative-sense, single-stranded RNA
- Similarity to Spooky virus: ~85% nucleotide identity, especially in conserved polymerase regions
- Reservoir host suspected: Native fruit bats (*Pteropus spp.*)
- Spillover origin: Likely zoonotic transmission event in Southeast Queensland, under investigation

3. Clinical Manifestations (SPOK-24)

- Incubation Period: Estimated 2-4 days
- Primary Symptoms (common): Fever, fatigue, rhinorrhea, non-productive cough, myalgia, sore throat

- Complications: Viral pneumonia, respiratory distress, neurological symptoms (rare), meningitis (<1%)
 - Mortality Rate: ~12% among confirmed hospitalized cases, higher risk in elderly, comorbid patients and children (<16 years old)
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4. Transmission and Epidemiology

- Primary Transmission: Respiratory droplets and close contact
 - Fomite-based transmission: Possible in healthcare and household settings
 - Notable Characteristics: Sustained human-to-human transmission; no confirmed long-distance airborne spread; clusters in hospitals and homes
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5. Diagnostic Considerations

- Confirmed by RT-PCR targeting the viral L gene (Doherty Institute primers)
 - Serological assays in development for retrospective surveillance
 - Clinical testing capacity currently limited in outbreak zones
 - Differential Diagnosis: Influenza, SARS-CoV-2, RSV, adenovirus, atypical pneumonia pathogens
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6. Nomenclature

- Proposed Virus Name: Spooky-like Virus 2024 (SPOK-24)
 - Proposed Disease Name: Spooky-Associated Respiratory Syndrome 2024 (SPOKARS-24)
 - Suggested Epidemic Name: 2024 Spooky-like Epidemic (SLE-24) or Australasian Respiratory Epidemic 2024 (ARE-24)
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7. Recommendations and Next Steps

- Continue genomic surveillance and viral characterization
- Update diagnostic protocols to include SPOK-24 in respiratory panels
- Support vaccine and therapeutic development targeting Spookyviruses
- Enhance international surveillance and harmonize case definitions
- Collaborate with veterinary/wildlife sectors to understand